

SWETHAMANI NAMANA

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PROFESSIONAL SUMMARY

Backend Developer with over **3 years** of experience building scalable **microservices** and **RESTful APIs** using **Java (Spring Boot)** and **Python (FastAPI, Flask)**. Skilled in designing, developing, and optimizing backend systems with **SQLAlchemy, JPA, and AWS cloud services**. Experienced in automating **CI/CD pipelines with GitHub Actions**, managing secure deployments using **AWS IAM and Secrets Manager**, and migrating legacy systems to cloud-native architectures. Improved API performance by **25%** through query optimization and **Redis caching**, delivering secure, high-performance backend solutions in Agile environments.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, GraphQL, HTML5, CSS3
- **Python Frameworks:** FastAPI, Flask, Uvicorn, SQLAlchemy, Pydantic, Marshmallow
- **Spring Frameworks:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, RESTful APIs
- **Cloud Platforms & Services:** Amazon Web Services (EC2, Lambda, S3, CodePipeline, IAM, Elastic Beanstalk, Secrets Manager, CloudWatch)
- **DevOps & CI/CD Tools:** GitHub Actions, Jenkins, Docker, SonarQube, Fortify, Blackduck, Veracode, Maven, Git
- **Databases:** MySQL, Redis, H2 Database, PostgreSQL
- **Monitoring & Logging:** ELK Stack (Elasticsearch, Logstash, Kibana, Rancher), Log4j, SLF4J, AWS CloudWatch, Dynatrace
- **Messaging Systems:** Apache Kafka
- **Testing & QA:** JUnit, Mockito, Postman, H2 Database, pytest
- **Version Control Systems:** Git, GitHub
- **API Tools & Standards:** Swagger, RESTful APIs, JSON, XML, OAuth2, JWT
- **AI/ML Technologies:** GPT-4, LangChain, Retrieval-Augmented Generation (RAG), Vector Databases

EXPERIENCE

Python Developer Truist Bank

August 2025-Present
Georgia, USA

- Designed and developed **Flask-based REST APIs** backed by **PostgreSQL** and **SQLAlchemy ORM**, powering secure data exchange between internal banking systems and improving API response time by 25%.
- Built APIs to handle **high-volume transaction processing**, using SQLAlchemy connection pooling, pagination, and optimized queries to ensure stable performance under heavy load.
- Implemented **reliability patterns** such as **circuit breakers, retries, and timeout handling** using Flask middleware and the requests library to minimize downtime during third-party service outages.
- Secured APIs with **JWT authentication, mutual TLS, and role-based authorization**, with secrets managed through **AWS Secrets Manager** to meet internal audit and data protection standards.
- Automated build and deployment pipelines using **GitHub Actions** and **AWS CodePipeline**, integrating **SonarQube** and **pytest** checks to maintain high-quality, secure code before release.
- Deployed **Dockerized services** on **AWS EC2** and **Lambda**, reducing deployment time by 50% and ensuring consistent release processes across environments.
- Set up **CloudWatch monitoring** and **ELK Stack dashboards** to track API performance, latency, and error trends, improving issue detection and resolution time.
- Collaborated with DevOps teams on **AWS environment setup**, network configurations, and deployment automation for production and non-production systems.
- Worked closely with fraud analytics and compliance teams to deliver APIs for **transaction validation, risk scoring, and audit logging**, improving data accuracy and fraud detection processes.
- Authored detailed **Swagger/OpenAPI documentation** and standardized endpoint design to streamline integrations across cross-functional teams

LLMOps QA Engineering Intern Tutorify AI

August 2024-Dec 2024
CT, USA

- Built an AI-powered quiz & assignment generator using **GPT-4 & LangChain** to automatically generate content from slides & notes.
- Developed a teaching assistant chatbot with **GPT-4, RAG pipeline, & Chroma vector DB** for real-time, context-aware query resolution.
- Designed a high-performance backend using **Python FastAPI & Uvicorn**, supporting low-latency AI workflows & seamless **API** integrations.
- Deployed the platform on **AWS EC2** for scalability & reliability in handling large datasets & high user interactions.
- Integrated AI tools with Canvas LMS via **RESTful APIs** to automate real-time syncing of quizzes, assignments, & grades.

- Streamlined backend workflows using **AWS Lambda** & automated **CI/CD** pipelines with **AWS CodePipeline** & **GitHub Actions** to ensure efficient deployment & cost optimization.
- Configured **AWS IAM** roles & **Secrets Manager** for secure credential management in **Python FastAPI** services, enhancing compliance with data protection standards.
- Optimized **FastAPI** performance with **Python** & **Redis** caching, reducing **API** response times by 15 percent under high user loads.

System Engineer

April 2021 – August 2023

Tata Consultancy Services-US Bank

Hyderabad, India

- Designed and deployed **RESTful APIs** using **Spring Boot** and **OAuth2**, improving secure access control and enabling seamless communication across financial systems.
- Migrated legacy **.NET** modules to Java-based microservices architecture, reducing technical debt and cutting deployment times by 30% through service modularization.
- Developed **GraphQL APIs** to simplify data aggregation across microservices and support flexible front-end queries
- Optimized backend response times by 35% by integrating **Redis** caching and rewriting inefficient SQL queries to improve application performance.
- Applied **multithreading, concurrency control, and async execution using CompletableFuture** to parallelize transaction processing.
- Utilized **Spring Cloud Resilience4J** for implementing circuit-breaker and fault-tolerance mechanisms across microservices, improving system reliability under high-load conditions.
- Developed and maintained **Kafka** pipelines handling approximately 50K messages/day, supporting asynchronous communication between critical microservices.
- Automated build and deployment workflows using **Jenkins** and **GitHub Actions**, enabling faster, error-free production releases and reducing manual handoffs.
- Increased test coverage from 60% to 90% using **JUnit** and **Mockito**, contributing to reduced defect leakage and more stable releases and addressing security vulnerabilities with **Twistlock** and **Fortify**.
- Documented key backend APIs using **Swagger/OpenAPI**, streamlining QA handoffs and simplifying backend–frontend collaboration across teams.
- Translated user stories into scalable backend modules by working closely with product managers, QA, and business analysts to ensure timely and high-quality delivery.
- Implemented **JWT** authentication and role-based authorization across internal services, strengthening data protection and access control.
- Utilized **Maven** to manage project dependencies and modular builds across services, enhancing build consistency and environment portability.
- Configured centralized logging using the **ELK Stack**, which helped accelerate incident diagnosis and ensured visibility across distributed systems.
- Conducted detailed peer code reviews with emphasis on performance, readability, and adherence to architectural standards to improve overall code quality.
- Assisted in triaging and resolving high-severity production issues, coordinating with DevOps and L2 support to restore service functionality quickly.
- Contributed to Agile ceremonies including planning, backlog grooming, and retrospectives, influencing sprint outcomes through proactive input.

CERTIFICATIONS

JPMorgan Chase Software Engineering Simulation on Forage

- Built & tested RESTful APIs using Spring Boot
- Integrated Kafka for message streaming & H2 as an in-memory database
- Debugged backend services & resolved application context & topic-related issues.

EDUCATION

Master of Science in Data Science

August 2023-Dec 2024

University of New Haven | West Haven, CT